

Prompt Contamination in Realistic Universal Cross-Domain Retrieval: Diagnosis and Text-Guided Rectification

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Abstract

Universal Cross-Domain Retrieval (UCDR) requires models to generalize to unseen domains and unseen categories at inference time. Recent prompt-learning methods for vision-language models have achieved strong performance on object-centric benchmarks, yet their behavior under cluttered multi-object scenes remains under-explored. In such scenes, co-occurring objects can act as contextual distractors and induce **prompt contamination**: visually salient distractors may dominate the candidate tokens used for instance-adaptive prompt generation, causing retrieval representations to drift away from the designated primary target. To diagnose this gap, we construct **DomainCOCO**, a Realistic UCDR benchmark with 180,000 images across six domains, where each sample contains a designated primary category and contextual distractors. DomainCOCO is intended as a controlled diagnostic benchmark for studying target-aware retrieval under multi-object clutter and stylized domain shifts. We further provide **Text-Guided Rectification Network (TGR-Net)**, a practical rectification baseline that uses text embeddings as semantic guidance to re-weight patch-level visual evidence before prompt generation. Its Candidate Prompt Purification (CPP) module reduces distractor-dominated prompt synthesis, while a teacher-student strategy with a learnable null token enables label-free inference. Multi-scale feature fusion and gallery-side refinement are used as optional inference components for accuracy-oriented deployment. Experiments on DomainNet and DomainCOCO show that TGR-Net provides consistent gains over strong CLIP-based baselines, with larger gains in the more cluttered DomainCOCO setting. Code is available at <https://github.com/YUEcjy13/TGR-Net>.

CCS Concepts

• Information systems → Retrieval models and ranking.

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Keywords

Universal Cross-Domain Retrieval, CLIP, Prompt Learning, Domain-COCO

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1 Introduction

Universal Cross-Domain Retrieval (UCDR) addresses visual retrieval under two simultaneous generalization requirements: the *query domain* and the *query categories* are both unseen at inference time [21, 28]. This setting is important for practical visual search systems, such as sketch-to-photo search, cross-domain product retrieval, and evolving retrieval engines, where both appearance distributions and category vocabularies may drift over time [40]. Compared with conventional cross-domain retrieval, UCDR requires handling *domain shift* and *category generalization* in a unified framework, making robust knowledge transfer a central challenge [30].

Large-scale vision-language models (VLMs), especially CLIP [24] and ALIGN [13], provide strong cross-modal priors for UCDR. Instead of full fine-tuning, which may distort pre-trained representations and incur high computational cost, recent methods increasingly adopt parameter-efficient prompt learning, including CoOp [42], VPT [14], and ProS [5]. A representative recent method, DePro [4], decouples prompts into domain-level and class-level components and synthesizes instance-adaptive class prompts from visual candidate tokens. However, the prevailing evaluation regime is still dominated by object-centric benchmarks, which may under-expose the contextual ambiguity encountered in practical retrieval scenarios.

Most existing UCDR benchmarks, such as DomainNet [22] and Sketchy [26], are largely *object-centric*: images often contain a single dominant foreground object and relatively limited background interference. We refer to this setting as **Conventional UCDR**. In contrast, real-world queries frequently appear as *multi-object scenes* with co-occurring categories, occlusion, cluttered background, and

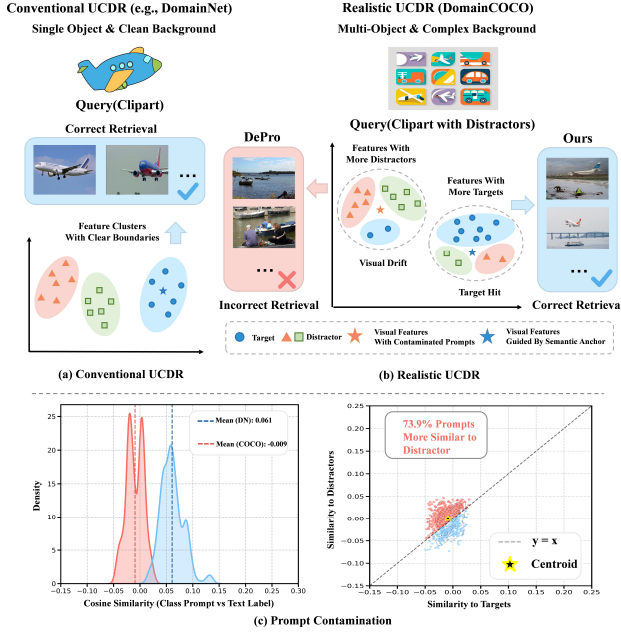


Figure 1: Conventional UCDR vs. Realistic UCDR. (a) Conventional UCDR benchmarks such as DomainNet are largely object-centric, where a single salient object often provides relatively clean retrieval evidence. (b) In Realistic UCDR, images contain multiple objects and cluttered context; candidate-token-driven class prompts may be dominated by visually salient distractors, leading to visual drift in the retrieval representation. (c) Evidence of prompt contamination: under clutter, instance-adaptive prompts become less aligned with the designated primary category and more aligned with co-occurring distractors, reducing retrieval reliability.

small targets [10, 17]. We refer to this setting as **Realistic UCDR**. This shift changes the evidence structure of each image: target-related and context-related evidence may coexist and compete in the same visual representation. As illustrated in Figure 1(a-b), Conventional UCDR tends to offer cleaner target evidence, whereas Realistic UCDR presents intertwined target and distractor cues, creating a systematic confounder for prompt-based retrieval models.

This reality gap exposes a failure mode that we term **prompt contamination**. Current prompt learners often synthesize instance-adaptive prompts by aggregating a small set of visual candidate tokens. This design implicitly assumes that these tokens are reliable evidence for the intended retrieval target. In Realistic UCDR, this assumption becomes fragile: visually salient tokens may originate from co-occurring objects or background clutter rather than the designated primary category [3]. As a result, the query representation may undergo **visual drift**, moving toward distractor-heavy evidence instead of the target category (Figure 1(b)). Figure 1(c) further shows that, under clutter, prompt-target alignment decreases while prompt-distractor alignment increases, making prompt contamination observable in the retrieval pipeline.

To diagnose this problem, we construct **DomainCOCO**, a Realistic UCDR benchmark derived from COCO [17] and LVIS [10]. DomainCOCO is designed as a controlled diagnostic benchmark rather than a replacement for naturally collected cross-domain data: each sample is associated with a designated primary category and co-occurring contextual distractors, while multiple stylized domains are generated with SDXL [23] to evaluate robustness under appearance shifts. This construction makes it possible to stress-test whether UCDR models retrieve the designated primary target or drift toward contextual co-occurrence.

We further provide the **Text-Guided Rectification Network (TGR-Net)** as a practical rectification baseline for Realistic UCDR. TGR-Net uses text embeddings as semantic guidance signals to re-weight patch-level visual evidence before it is consumed by the prompt generator. Its key component, **Candidate Prompt Purification (CPP)**, provides rectified tokens to the adaptive prompt generator, thereby reducing the dominance of distractor-driven evidence. To enable label-free inference, we adopt a teacher-student strategy: the teacher uses class names during training to provide semantic guidance, while the student learns a class-agnostic structural prior through a learnable **null token** for deployment without test labels. In addition, multi-scale feature fusion and gallery-side refinement are used as optional inference components to improve robustness under scale variation and query-gallery mismatch.

Experiments on both Conventional UCDR and DomainCOCO show that TGR-Net provides consistent gains over strong CLIP-based baselines. More importantly, the results provide diagnostic evidence that prompt contamination becomes more harmful in cluttered multi-object scenes, and that regulating the visual evidence used for prompt generation can reduce this failure mode.

Our main contributions are summarized as follows:

- **A Realistic UCDR Diagnostic Benchmark:** We construct **DomainCOCO** to stress-test UCDR models under multi-object composition, contextual distractors, and stylized domain shifts.
- **A Prompt Contamination Diagnosis:** We identify and analyze prompt contamination as a failure mode of candidate-token-driven prompt learning in cluttered retrieval scenes.
- **A Practical Rectification Baseline:** We provide TGR-Net as an initial baseline for Realistic UCDR, using semantic guidance and teacher-student distillation to reduce distractor-dominated prompt generation while retaining label-free inference.
- **A Practical Evaluation Study:** We evaluate TGR-Net on both object-centric and realistic UCDR settings, with ablations and qualitative analyses that clarify the role of rectification, multi-scale inference, and gallery-side refinement.

2 Related Work

Universal Cross-Domain Retrieval (UCDR). Cross-Domain Retrieval (CDR) studies retrieval under distribution shifts between heterogeneous visual domains [6, 11, 20, 30]. Early work mainly considered *closed-set* transfer via invariant feature learning [7, 19] or deep hashing [18, 29, 33]. UCDR is more demanding because both the query domain and the query category are unseen at inference

time; Paul et al. [21] formalized this setting with semantic propagation (SnMpNet), and later methods exploit structural invariants [28] or spatial-contextual cues such as SCNNNet [1]. With large-scale VLMs such as CLIP and ALIGN [13, 24], prompt-based adaptation has become a strong low-cost paradigm for UCDR. Recent methods such as ProS [5] and DePro [4] improve prompt transfer on standard benchmarks; DePro further decouples domain and class prompts and synthesizes instance-adaptive class prompts from visual candidates. Our work builds on this line but studies a different stress factor: whether candidate-token prompt synthesis remains reliable when the image contains a designated target together with strong contextual distractors. Beyond retrieval, multi-view representation learning has explored complementary evidence preservation through structure-aware conditional generation, shared-space feature aggregation, diffusion-contrastive recovery, and global–local cross-view sequence modeling [35–38]. These methods address clustering rather than UCDR, but provide a related perspective on avoiding indiscriminate fusion of heterogeneous evidence.

Prompt Learning for Frozen Vision–Language Models. Prompt learning adapts frozen VLMs by optimizing lightweight parameters instead of fully fine-tuning the backbone [40]. Representative approaches include text-side prompts (CoOp/CoCoOp) [41, 42], visual-side prompts (VPT) [14], and deep multi-modal prompting (MaPLE) [15]. Further improvements come from gradient alignment [43], test-time adaptation [16, 27], adapter-style parameterization [8, 9, 34], and LLM-assisted prompt synthesis [31, 44]. These methods are effective, but instance-adaptive prompt synthesis often assumes that salient visual tokens are useful evidence for the intended target. In cluttered multi-object scenes, saliency may instead come from co-occurring objects or background regions [3], causing what we call **prompt contamination**. TGR-Net is therefore positioned as an initial rectification baseline: it regularizes candidate evidence before prompt generation rather than replacing the underlying UCDR prompting framework.

Benchmarks: From Object-Centric to Diagnostic Realism. Benchmarks have shaped domain adaptation and retrieval research, from Office-31 [25] to UCDR datasets such as DomainNet [22], Sketchy [26], and TU-Berlin [2]. However, these benchmarks are largely object-centric, so a single foreground object often provides clean retrieval evidence. Real retrieval scenarios more frequently involve multi-object scenes, small targets, occlusion, and spurious context, as reflected in COCO [17] and LVIS [10]; such context can affect robustness and bias [32, 39]. DomainCOCO is designed as a *diagnostic* benchmark rather than a replacement for naturally collected data: by explicitly recording a primary target and contextual distractors, it stress-tests whether UCDR models retrieve the designated category or drift toward co-occurring context.

3 The DomainCOCO Benchmark

DomainCOCO is a **Realistic UCDR** benchmark designed to diagnose retrieval failures under cluttered, multi-object compositions [4, 5]. Unlike traditional object-centric datasets that mainly feature isolated objects, DomainCOCO starts from MS-COCO [17] and LVIS [10], whose images contain complex scene layouts with multiple annotated categories. For each scene, we designate one

category as the *primary* retrieval target, while the remaining co-occurring categories are treated as **contextual distractors**. We then extend each selected scene into the *Real* domain and five stylized variants, enabling a controlled analysis of domain and category shifts while approximately preserving the underlying scene composition. This design makes DomainCOCO a diagnostic testbed for analyzing whether retrieval models remain aligned with the designated primary category or drift toward contextual distractors.

3.1 Construction Pipeline

As shown in Figure 2, DomainCOCO is curated through a multi-stage pipeline that preserves scene complexity while introducing stylistic diversity.

Category Selection and Multi-object Filtering. We select 150 foreground categories from the LVIS vocabulary, prioritized by their total instance frequency to ensure sufficient data density [10]. To enforce Realistic UCDR conditions, we filter the source data to retain images containing at least two distinct annotated categories. For each retained image, the primary class is recorded as the retrieval target, while the co-occurring annotated categories are retained as contextual distractor metadata. This metadata supports controlled analysis of contextual confounding, including co-occurrence patterns and approximate spatial relations, without requiring additional supervision during model training.

Generative Domain Stylization. To construct five stylized domains (*Clipart*, *Infograph*, *Sketch*, *Painting*, *Quickdraw*) in addition to the *Real* domain, we employ Stable Diffusion XL (SDXL) [23] within a ComfyUI-based generation framework. For each target domain, representative open-source LoRA adapters are used to increase stylistic diversity and reduce reliance on a single visual style. The generation is governed by a structured prompt:

$$P = [P_{\text{domain}}] + [C_{\text{primary}}]^{++} + [C_{\text{distractors}}] + [P_{\text{negative}}], \quad (1)$$

where $[P_{\text{domain}}]$ specifies the target style, $[C_{\text{primary}}]^{++}$ emphasizes the designated primary category, $[C_{\text{distractors}}]$ encourages preservation of co-occurring contextual categories, and $[P_{\text{negative}}]$ suppresses common generative artifacts. This formulation aims to approximately preserve scene semantics and distractor structure across domains, while we acknowledge that generative stylization may still introduce artifacts or imperfect object preservation.

Verification-oriented Curation. Generative stylization may produce hallucinations, missing objects, or distorted scene layouts. We therefore combine automatic filtering with human verification. For each class–domain pair, we first generate 300 candidate images and continue verification until 200 valid samples are retained, yielding **180,000 images** in total (150 categories $\times$ 200 images $\times$ 6 domains). Human verification follows three criteria: identifiable primary target, at least one recognizable contextual distractor, and approximately preserved scene composition. As summarized in Table 1, three annotation groups are used; each group handles two domains with two initial annotators and one adjudicator. The initial agreement rate is 95.7%, and the remaining 4.3% disagreements are resolved by the adjudicator. Invalid generations rejected during verification account for 6.9% of the final retained set; the remaining

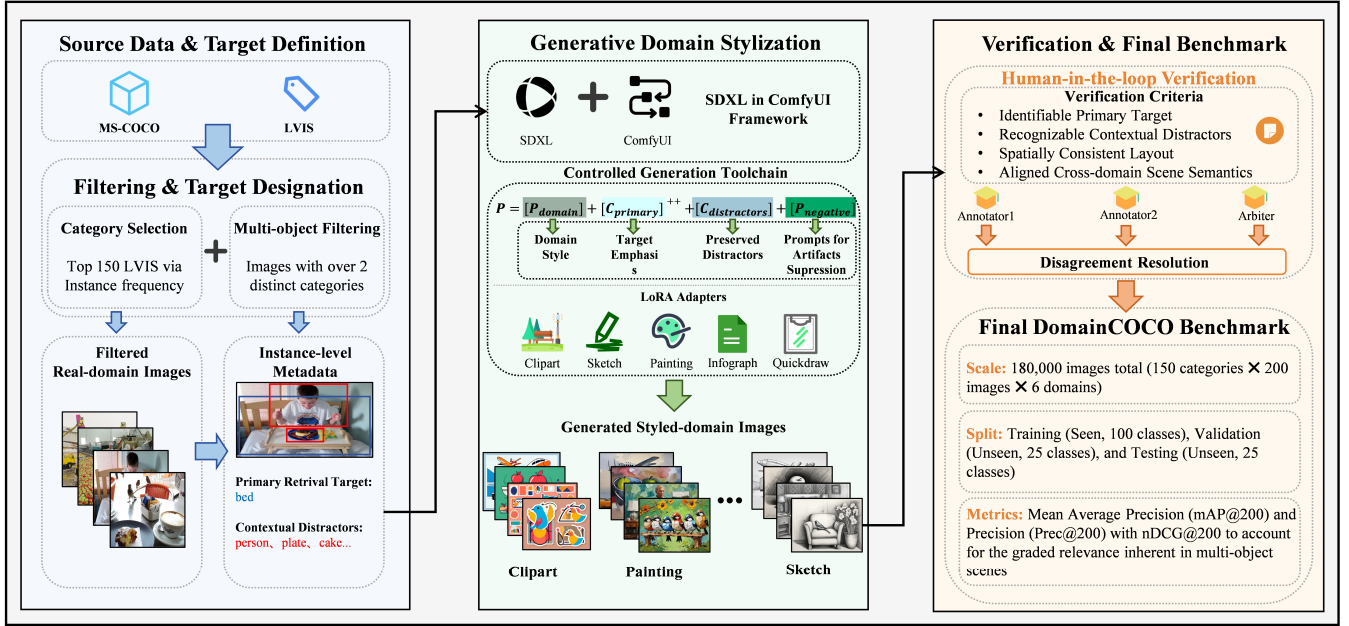


Figure 2: Construction pipeline of DomainCOCO. DomainCOCO is built from MS-COCO and LVIS through three stages: source data filtering and primary-target designation, SDXL-based generative domain stylization with structured prompts and LoRA adapters, and verification-oriented curation before forming the final benchmark. The pipeline records both the primary retrieval target and contextual distractors, enabling controlled evaluation of target-aware retrieval under multi-object clutter.

Table 1: DomainCOCO quality-control summary. The table reports curation checks that are not already captured by the construction pipeline in Figure 2.

Item	Protocol / statistic
Candidate generation	300 candidates per class-domain pair
Valid retained samples	200 per class-domain pair; 180,000 images total
Human verification	3 groups; each group checks 2 domains
Decision protocol	2 initial annotators; disagreements resolved by a 3rd adjudicator
Agreement rate	95.7% initial agreement; 4.3% adjudicated
Filtering rate	Invalid rejected generations equal 6.9% of the final retained set; unused candidates serve as reserves

generated candidates serve as reserves for completing the 200 valid samples per class-domain pair.

3.2 Evaluation Protocol and Metrics

Splits. Following standard UCDR protocols [21, 28], we partition the 150 categories into three disjoint sets: **Training (Seen, 100 classes)**, **Validation (Unseen, 25 classes)**, and **Testing (Unseen, 25 classes)**. Category generalization is measured by evaluating models only on the *Unseen* category sets.

Metrics. We adopt Mean Average Precision (mAP@200) and Precision (Prec@200), consistent with prior UCDR literature [21, 28].

For DomainCOCO, we additionally report **nDCG@200** [12] to account for graded relevance in multi-object scenes. Specifically, a retrieved image receives $rel = 1.0$ if it matches the designated primary target, $rel = 0.5$ if it shares contextual distractors but misses the primary target, and $rel = 0.0$ otherwise. The middle weight is used only to distinguish context-related failures from completely irrelevant retrievals; mAP@200 and Prec@200 remain the primary target-centric metrics and are used for target-correctness claims. Formally, for a query q and ranking list K :

$$DCG@K = \sum_{i=1}^K \frac{2^{rel_i} - 1}{\log_2(i + 1)}, \quad (2)$$

where nDCG@K is obtained by normalizing the DCG value with the Ideal DCG (IDCG).

3.3 Release, Compliance, and Scope

Open Science and Reproducibility. We will release the **DomainCOCO Toolkit**, including structured prompt templates, ComfyUI workflows, LoRA adapter references, and annotation manifests.

Compliance and Licensing. DomainCOCO follows the usage policies of MS-COCO [17] and LVIS [10]. For compliance, we distribute annotation manifests and image retrieval or generation scripts rather than redistributing raw source images.

Scope of the Benchmark. DomainCOCO is intended as a controlled diagnostic benchmark for target-centric retrieval under contextual interference. It does not aim to cover all forms of user intent

in multi-object retrieval, such as queries involving multiple interacting objects. Instead, by explicitly designating a primary category and recording contextual distractors, it provides a reproducible setting for studying how clutter and co-occurrence affect UCDR models.

4 Methodology

We instantiate the **Text-Guided Rectification Network (TGR-Net)** as a practical baseline for reducing prompt contamination in multi-object retrieval scenes. As Figure 4 illustrates, the same category may appear under diverse stylized domains and cluttered compositions, making candidate evidence less reliable for prompt generation. The goal is not fully supervised target localization, but regularizing the visual evidence used for prompt generation so that salient distractors are less likely to dominate instance-adaptive prompts. As shown in Figure 3, TGR-Net contains two main components: (1) a **Text-Guided Rectification (TGR)** module that re-weights patch embeddings using semantic guidance, and (2) **Candidate Prompt Purification (CPP)**, which feeds rectified visual candidates to the prompt generator. To support label-free deployment, we distill the teacher’s semantic-guided rectification behavior into a student model with a learnable **null token**.

4.1 Preliminaries: Decoupled Prompting

We follow the decoupled prompting paradigm [4], adapting a frozen CLIP [24] backbone (image encoder $\mathcal{V}$ and text encoder $\mathcal{T}$) with two prompt components:

- **Universal Domain Prompts (UDPs):** A set of learnable vectors $\mathbf{P}_{dom} \in \mathbb{R}^{M_d \times D}$ shared across samples to bridge global domain shifts.
- **Class Prompts (CPs):** Instance-adaptive vectors $\mathbf{P}_{cls}(I) \in \mathbb{R}^{M_c \times D}$ generated to capture category-related visual semantics.

The synthesis of CPs is governed by an **Adaptive Semantic Prompter (ASP)**. Following DePro [4], ASP is implemented as a lightweight transformer module with l layers. It takes visual candidate tokens and meta-class prompts as input and transforms them into sample-specific prompt vectors. In Realistic UCDR, however, the visual tokens consumed by ASP may contain cluttered evidence from co-occurring objects or background regions [3], leading to visual drift. TGR-Net therefore rectifies this input stream before prompt generation.

4.2 Text-Guided Rectification (TGR)

The TGR module uses a semantic or structural anchor to re-weight visual evidence before it is passed to ASP.

Anchor Modes: Semantic vs. Structural. We define an anchor query $\mathbf{q}$ in two modes to bridge training and inference:

$$\mathbf{q} = \begin{cases} \mathbf{t}_{gt} = \mathcal{T}(\text{"a photo of [CLASS]", } \mathbf{P}_{dom}), & \text{Teacher (train)} \\ \mathbf{t}_\emptyset \in \mathbb{R}^D, & \text{Student (test)} \end{cases} \quad (3)$$

where $\mathbf{t}_{gt}$ provides explicit category semantics during training. At inference time, labels are unavailable, so the student uses a learnable **null token** $\mathbf{t}_\emptyset$ shared across images. Through distillation, $\mathbf{t}_\emptyset$ learns

a class-agnostic structural prior that tends to emphasize broadly informative foreground evidence and reduce the dominance of background clutter.

Anchor-Guided Attention. Let $\mathbf{F}_{raw}$ denote the ViT patch embeddings [24]. We compute an attention map $\mathbf{A}$ as:

$$\mathbf{A}_i = \frac{\exp(\mathbf{q}^\top (W_K \mathbf{f}_i) / \sqrt{D})}{\sum_{j=1}^L \exp(\mathbf{q}^\top (W_K \mathbf{f}_j) / \sqrt{D})}. \quad (4)$$

For the teacher, $\mathbf{A}$ acts as semantic guidance; for the student, it acts as a learned structural weighting over visual tokens.

Gated Global Rectification. We aggregate anchor-aligned visual evidence and inject it back into the token sequence through a gated residual connection:

$$\mathbf{F}_{rect} = \mathbf{F}_{raw} + \gamma \cdot \text{MLP} \left(\sum_{i=1}^L \mathbf{A}_i (W_V \mathbf{f}_i) \right), \quad (5)$$

where γ is initialized to 0. This zero-initialization preserves the CLIP feature manifold early in training and introduces rectified evidence gradually, encouraging the representation to down-weight distractor-dominated evidence while retaining visual semantics.

4.3 Candidate Prompt Purification (CPP)

CPP reduces prompt contamination by replacing raw candidate tokens with rectified tokens before prompt synthesis:

$$\mathbf{P}_{cls}^{clean} = \text{ASP}(\mathbf{F}_{rect}). \quad (6)$$

Rather than claiming explicit object localization, CPP encourages ASP to synthesize class prompts from less contaminated visual evidence. The resulting prompts are combined with UDPs to form the final prompted representation.

4.4 Joint Semantic Alignment

TGR-Net learns a retrieval space where rectified image features are aligned with category text features. Following the global alignment strategy in DePro [4], we optimize the model using the following components.

Feature Projection. For an image I , the visual representation $\mathbf{z}_\emptyset$ is obtained by feeding the rectified tokens $\mathbf{F}_{rect}$ and prompts $(\mathbf{P}_{dom}, \mathbf{P}_{cls}^{clean})$ into the frozen image encoder $\mathcal{V}$. The text representation $\mathbf{z}_t$ is generated by the text encoder $\mathcal{T}$ using category labels wrapped in a prompt template.

Alignment Objectives. We minimize the following losses:

- **Image–Text Matching ($\mathcal{L}_{itm}$):** A contrastive objective [24] that aligns matched image–text pairs and separates mismatched pairs in a batch.
- **Domain-aware Triplet Loss ($\mathcal{L}_{dtr}$):** This loss pulls same-category samples across domains closer and pushes different-category samples apart, promoting domain-invariant category structure.
- **Prompt Decoupling ($\mathcal{L}_{itm}^{dec}$):** Following DePro [4], this auxiliary objective encourages UDPs and CPs to preserve their intended roles for domain-level and category-level adaptation.

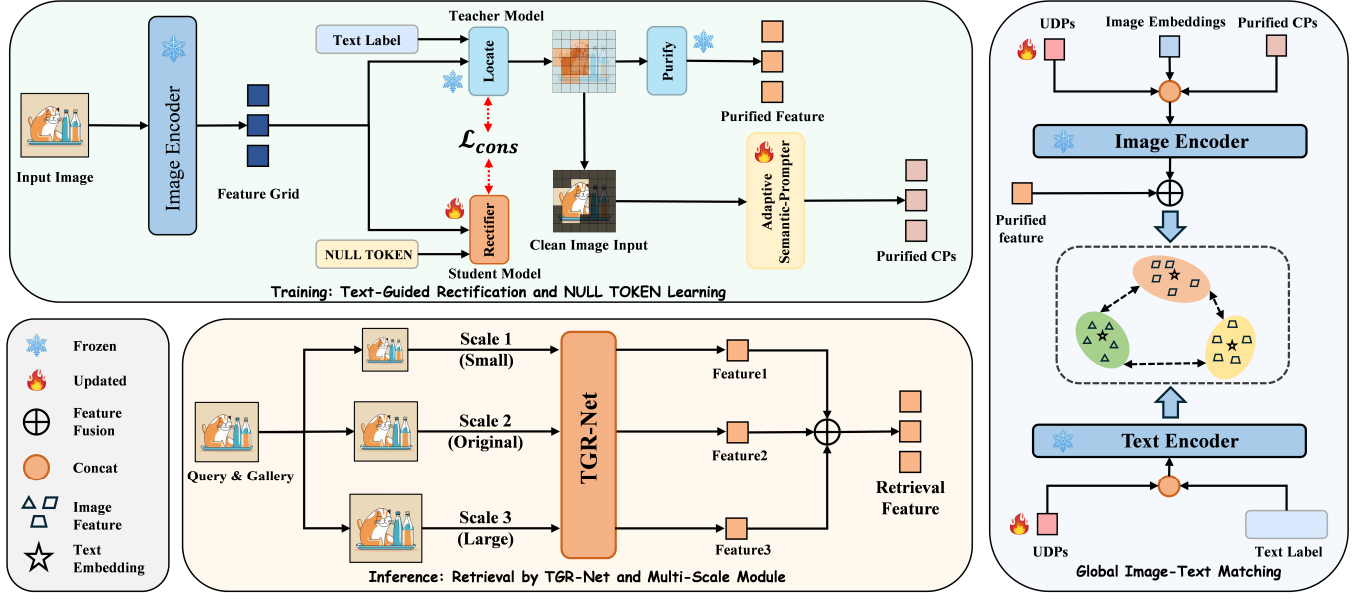


Figure 3: The overall framework of TGR-Net. (Top-left) Joint Training Stage: The teacher uses class labels as semantic guidance for token rectification, while the student learns a class-agnostic structural prior through a learnable null token. Both branches are aligned by a consistency loss $\mathcal{L}_{cons}$. (Bottom-left) Inference Stage: The student performs label-free rectification. Multi-Scale Feature Fusion (MSF) and gallery refinement are optional inference components for accuracy-oriented deployment. (Right) Joint Semantic Alignment: Rectified class prompts (CPs) and universal domain prompts (UDPs) are combined to align image and text representations in a shared retrieval space.

4.5 Teacher-Student Consistency

To enable label-free inference, we distill rectification behavior from the teacher to the student. Let $(A_{tea}, F_{rect}^{tea})$ and $(A_{stu}, F_{rect}^{stu})$ denote the attention maps and rectified tokens from the teacher and student, respectively. The student is optimized with:

$$\mathcal{L}_{cons} = \|A_{stu} - A_{tea}\|_2^2 + \lambda_{feat} \|F_{rect}^{stu} - F_{rect}^{tea}\|_2^2. \quad (7)$$

This objective makes the null token approximate the teacher’s semantic-guided weighting behavior without labels.

4.6 Inference Strategy

Multi-Scale Feature Fusion (MSF). To handle scale variation, we optionally fuse features from three scales $s \in \{I_{small}, I_{orig}, I_{large}\}$ with a fusion coefficient β :

$$f^{(s)} = (1 - \beta) \cdot \text{Pool}(F_{raw}^{(s)}) + \beta \cdot \text{Pool}(F_{rect}^{(s)}). \quad (8)$$

The final representation f_{final} concatenates features across scales. MSF is an accuracy-oriented option and can be disabled when inference efficiency is prioritized.

Training Objective. The student model is optimized by:

$$\mathcal{L}_{total} = \underbrace{\mathcal{L}_{itm} + \mathcal{L}_{dtr} + \mathcal{L}_{itm}^{dec}}_{\mathcal{L}_{DePro}} + \mathcal{L}_{reg} + \lambda_{cons} \mathcal{L}_{cons}. \quad (9)$$

This objective preserves the decoupled prompt learning paradigm while adding a label-free rectification prior for Realistic UCDR.

5 Experiments

In this section, we evaluate TGR-Net under standard UCDR protocols [4, 21, 28]. The experiments serve two purposes: diagnosing whether DomainCOCO exposes clutter-related failures that object-centric benchmarks under-emphasize, and testing whether a practical rectification baseline can reduce this failure mode. We report results on conventional DomainNet and realistic DomainCOCO, followed by ablations, sensitivity analysis, and qualitative diagnostics.

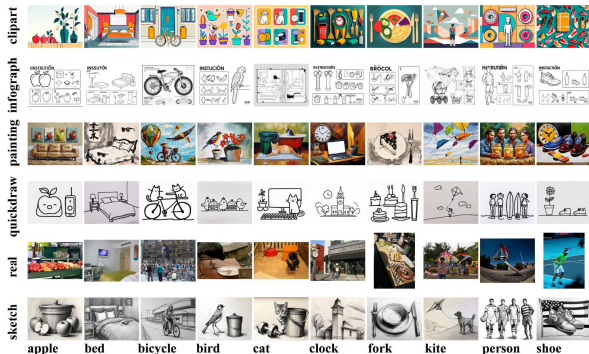


Figure 4: Examples from DomainCOCO. Samples from six domains across representative categories, illustrating stylized domain shifts and cluttered multi-object scenes.

Table 2: Detailed results on DomainNet under UCDR. DomainNet is used as a conventional object-centric benchmark to verify that the rectification baseline remains compatible with standard UCDR evaluation. We report all five query-domain splits under Unseen and Mixed Gallery settings. ★: traditional methods; †: CLIP-based methods.

Training Domains	Query	Method	Venue	Unseen Gallery		Mixed Gallery	
				mAP@200	Prec@200	mAP@200	Prec@200
Real, Infograph, Quickdraw, Painting, Clipart	Sketch	SnMpNet★	ICCV'21	28.45	23.12	24.87	20.15
		SASA★	SIGIR'22	49.88	42.36	44.92	38.11
		CLIP (ZS)†	ICML'21	40.12	33.58	34.75	28.42
		VPT†	ECCV'22	59.25	54.12	53.48	48.95
		ProS†	CVPR'24	61.34	57.05	55.62	51.87
		DePro†	SIGIR'25	65.91	61.98	58.95	55.57
		TGR-Net	Ours	67.49	63.82	60.41	57.46
Real, Sketch, Quickdraw, Painting, Clipart	Infograph	SnMpNet★	ICCV'21	18.75	15.68	16.32	13.54
		SASA★	SIGIR'22	25.43	22.05	22.58	19.12
		CLIP (ZS)†	ICML'21	45.68	40.85	39.95	35.42
		VPT†	ECCV'22	49.52	45.16	44.52	40.75
		ProS†	CVPR'24	52.85	49.62	47.65	45.18
		DePro†	SIGIR'25	56.71	53.94	49.79	47.51
		TGR-Net	Ours	58.75	56.24	51.84	49.77
Real, Sketch, Infograph, Painting, Clipart	Quickdraw	SnMpNet★	ICCV'21	16.12	11.95	14.05	10.32
		SASA★	SIGIR'22	23.85	18.34	19.68	15.35
		CLIP (ZS)†	ICML'21	6.95	5.23	5.58	2.95
		VPT†	ECCV'22	22.85	19.45	18.15	15.68
		ProS†	CVPR'24	26.42	23.75	21.56	19.82
		DePro†	SIGIR'25	29.48	27.74	22.90	21.84
		TGR-Net	Ours	30.18	28.54	23.35	22.59
Real, Sketch, Infograph, Quickdraw, Clipart	Painting	SnMpNet★	ICCV'21	39.42	32.55	35.54	29.54
		SASA★	SIGIR'22	57.65	50.75	53.38	46.95
		CLIP (ZS)†	ICML'21	60.15	53.85	55.20	49.05
		VPT†	ECCV'22	69.85	63.58	65.92	60.12
		ProS†	CVPR'24	73.45	67.92	69.58	64.65
		DePro†	SIGIR'25	75.19	70.07	70.30	65.73
		TGR-Net	Ours	76.11	71.00	71.34	66.80
Real, Sketch, Infograph, Quickdraw, Painting	Clipart	SnMpNet★	ICCV'21	40.85	32.35	36.65	28.85
		SASA★	SIGIR'22	42.75	35.75	38.45	32.12
		CLIP (ZS)†	ICML'21	58.85	50.12	54.68	45.75
		VPT†	ECCV'22	71.55	66.25	67.82	62.58
		ProS†	CVPR'24	74.52	70.15	70.45	66.52
		DePro†	SIGIR'25	77.14	73.18	71.68	68.12
		TGR-Net	Ours	78.59	74.95	73.38	70.15

5.1 Datasets and Metrics

Datasets. We evaluate on:

- **DomainNet [22]:** ~0.6M images from six domains (*Real*, *Sketch*, *Infograph*, *Painting*, *Clipart*, *Quickdraw*). We adopt

Table 3: Comparison with SOTA methods on DomainCOCO under UCDR. We report nDCG@200, mAP@200, and Prec@200. TGR-Net provides consistent gains over strong prompt-learning baselines.

Training Domains	Query	Method	Venue	Unseen Gallery			Mixed Gallery		
				nDCG@200	mAP@200	Prec@200	nDCG@200	mAP@200	Prec@200
Real, Quickdraw, Infograph, Painting, Clipart	Sketch	CLIP (ZS) [†]	ICML'21	25.43	20.12	15.54	22.18	18.25	13.15
		VPT [†]	ECCV'22	34.25	28.56	21.48	31.54	25.85	19.23
		ProS [†]	CVPR'24	39.87	34.25	26.54	37.15	31.45	23.85
		DePro [†]	SIGIR'25	43.15	37.91	29.48	41.32	31.32	24.32
		TGR-Net	Ours	46.40	40.46	31.98	44.67	33.73	26.72
Real, Quickdraw, Infograph, Painting, Clipart	Infograph	CLIP (ZS) [†]	ICML'21	22.35	17.58	13.45	19.45	15.36	11.24
		VPT [†]	ECCV'22	31.18	25.65	19.34	28.75	23.36	17.15
		ProS [†]	CVPR'24	36.85	30.94	23.68	33.95	28.16	21.45
		DePro [†]	SIGIR'25	46.24	40.25	31.22	44.72	33.59	26.09
		TGR-Net	Ours	50.60	43.84	34.78	49.19	37.01	29.48
Real, Quickdraw, Infograph, Painting, Clipart	Quickdraw	CLIP (ZS) [†]	ICML'21	18.15	14.28	10.85	15.26	12.18	9.15
		VPT [†]	ECCV'22	26.35	22.18	16.94	23.58	19.26	14.55
		ProS [†]	CVPR'24	31.76	26.85	20.45	28.95	23.94	18.15
		DePro [†]	SIGIR'25	44.55	38.66	30.33	42.76	31.81	25.10
		TGR-Net	Ours	47.40	41.16	32.57	45.74	34.25	27.28
Real, Quickdraw, Infograph, Painting, Clipart	Painting	CLIP (ZS) [†]	ICML'21	29.85	24.38	18.94	26.45	21.46	16.14
		VPT [†]	ECCV'22	38.76	32.58	25.46	35.65	29.45	22.35
		ProS [†]	CVPR'24	44.95	38.46	30.45	41.56	35.24	27.46
		DePro [†]	SIGIR'25	46.50	40.63	31.57	44.98	33.76	26.31
		TGR-Net	Ours	49.19	42.71	33.54	47.90	35.64	28.35
Real, Quickdraw, Infograph, Painting, Clipart	Clipart	CLIP (ZS) [†]	ICML'21	28.46	23.18	17.65	25.16	20.58	15.26
		VPT [†]	ECCV'22	37.26	31.38	24.45	34.56	28.46	21.64
		ProS [†]	CVPR'24	42.96	36.85	28.64	40.16	33.84	25.86
		DePro [†]	SIGIR'25	46.49	39.68	31.20	44.85	33.08	25.96
		TGR-Net	Ours	49.32	41.67	33.48	47.72	34.75	27.08

the standard UCDR split and protocol [4, 21, 28] (245/55/45 train/val/test classes).

- **DomainCOCO (Ours):** A diagnostic benchmark built from MS-COCO/LVIS with multi-object scenes, primary targets, contextual distractors, and stylized domains. We use a strict **100/25/25 (Seen/Val/Unseen)** class split and report results only on Unseen classes.

Evaluation Metrics. We report **mAP@200** and **Prec@200** on both benchmarks. DomainCOCO also uses **nDCG@200** for graded relevance. All results are averaged over 3 random seeds.

5.2 Implementation Details

We use CLIP ViT-B/32 [24] as the frozen image/text encoders with the default 224×224 input resolution. We strictly follow the UCDR training protocol: models are trained only on Seen classes and never access Unseen test-domain images during training. On DomainCOCO, supervision uses only category names as text prompts (no extra annotations). Training is conducted on a single **NVIDIA A100 GPU** using Adam (lr 1×10^{-4} , batch size 60) with cosine decay and warm-up. Unless stated otherwise, we set $\beta = 0.3$ and $\beta_{gal} = 0.3$.

5.3 Comparison with the State-of-the-Arts

We compare TGR-Net with state-of-the-art (SOTA) UCDR methods categorized into: (i) **Traditional methods:** SnMpNet [21] and SASA [28]; (ii) **CLIP-based methods:** zero-shot CLIP [24], VPT [14], ProS [5], and the baseline **DePro** [4].

5.3.1 Results on DomainNet. Table 2 reports detailed results on DomainNet, our conventional object-centric reference setting. CLIP-based prompt learners outperform traditional methods, consistent with prior UCDR work [4]. TGR-Net gives modest but consistent gains over DePro across query domains, suggesting that rectification remains compatible with standard UCDR. Since DomainNet contains cleaner object evidence than DomainCOCO, we treat it as a sanity check rather than the main source of our diagnostic claim.

5.3.2 Results on DomainCOCO (Realistic UCDR). DomainCOCO evaluates models under explicit multi-object composition and contextual distractors. Table 3 supports two observations. ① **Diagnostic finding.** Compared with object-centric DomainNet, DomainCOCO produces a harder target-centric retrieval problem: strong prompt learners such as DePro remain competitive but show larger room for improvement under clutter and mixed-gallery retrieval. This suggests that candidate-token prompt synthesis can be fragile when salient distractors compete with the intended category, supporting DomainCOCO as a stress test for prompt contamination

rather than merely another stylized benchmark. ② **Rectification baseline.** TGR-Net provides consistent gains over DePro on Unseen Gallery, with average improvements of **+3.2 nDCG**, **+2.5 mAP**, and **+2.5 Prec**. These gains indicate that regulating candidate evidence before prompt generation is a useful first step, while leaving room for future target-aware retrieval methods under clutter.

5.4 Ablation Studies

Component-wise Effectiveness. We perform progressive ablations on DomainCOCO with **Clipart** queries, starting from **DePro** and incrementally enabling the components of our practical baseline.

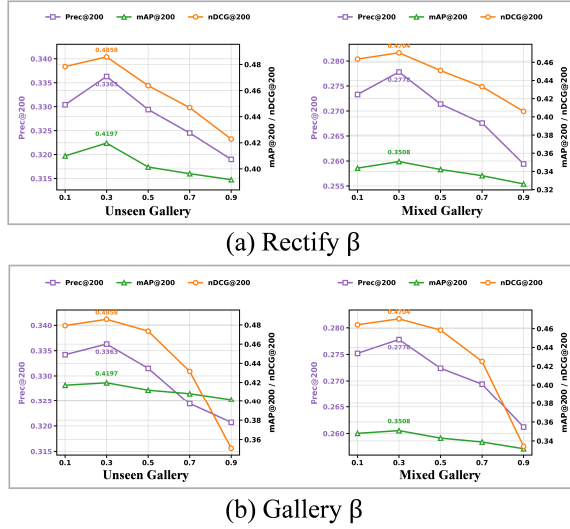


Figure 5: Hyperparameter Sensitivity on DomainCOCO. (a) Impact of rectification coefficient β . (b) Impact of gallery refinement coefficient β_{gal} . Performance is stable within a broad range, with the best setting around $\beta = 0.3$ and $\beta_{gal} = 0.3$.

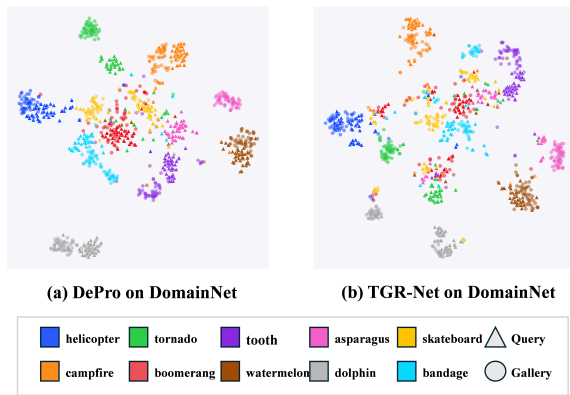


Figure 6: t-SNE on DomainNet. Comparison between (a) DePro and (b) TGR-Net. TGR-Net gives slightly more compact clusters and clearer cross-domain alignment.

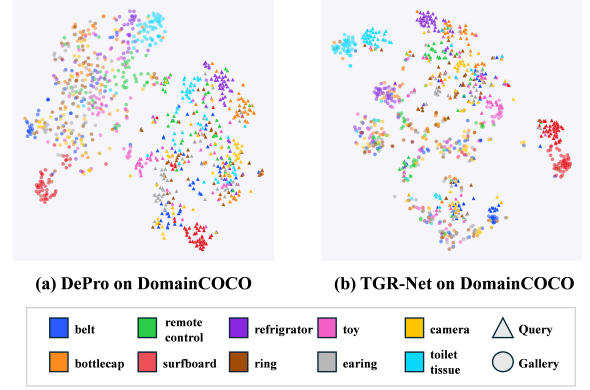


Figure 7: t-SNE on DomainCOCO. TGR-Net produces clearer category structures and fewer background-induced confusions.

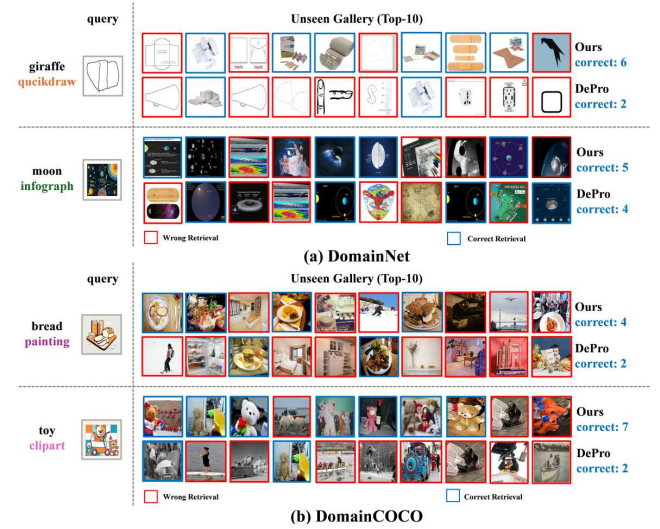


Figure 8: Top-10 Retrieval Comparison. Qualitative comparison between DePro and our TGR-Net on (a) DomainNet and (b) DomainCOCO. Blue borders indicate correct retrievals, while red borders indicate wrong ones. TGR-Net retrieves more target-consistent images in abstract sketches and multi-object DomainCOCO scenes.

Table 4 reports results under both Unseen and Mixed Gallery settings.

We observe that:

- **+MSF**: provides a small but consistent gain (+0.25 nDCG in Unseen), reflecting the benefit of multi-scale cues in cluttered scenes.
- **+Text-Guided Rectify**: yields a clear boost (+0.85 nDCG and +0.64 mAP over MSF in Unseen) by improving target-consistent token selection under clutter.
- **+Text-Guided Gallery**: further improves query-gallery alignment (e.g., +0.68 nDCG in Unseen).

Table 4: Component-wise effectiveness analysis on DomainCOCO (Query: Clipart). We report nDCG@200, mAP@200, and Prec@200 under both Unseen and Mixed Gallery settings. The baseline is the recent SOTA method DePro [4].

Method	Unseen Gallery			Mixed Gallery		
	nDCG	mAP	Prec	nDCG	mAP	Prec
Baseline (DePro)	46.49	39.68	31.20	44.85	33.08	25.96
+ MSF	46.74	39.79	31.43	45.13	33.14	26.05
+ Text-Guided (Rectify)	47.59	40.43	32.19	46.05	33.78	26.43
+ Text-Guided (Gallery)	48.27	40.92	32.74	46.85	34.19	26.64
+ CPP (Full TGR-Net)	49.32	41.67	33.48	47.72	34.75	27.08

- **+CPP (Full):** gives the strongest result in this baseline family by feeding rectified candidates into the prompt generator and reducing prompt contamination.

5.5 Hyperparameter Sensitivity Analysis

Figure 5 shows stable performance for $\beta, \beta_{gal} \in [0.2, 0.4]$, with the best results around 0.3. Very small coefficients underuse rectified evidence, while overly large ones make the representation depend too strongly on the correction branch. This supports the intended calibration role: rectification should guide candidate evidence, not replace the original CLIP representation.

5.6 Visualization

Figures 6–7 visualize feature distributions on DomainNet and DomainCOCO. The DomainNet difference is moderate, while DomainCOCO shows clearer category structures and fewer background-induced confusions, consistent with the view that rectification is more useful when target and distractor evidence are entangled. Figure 8 further shows more target-consistent Top-10 retrievals. These observations are not standalone localization evidence, but they agree with the ranking gains in Tables 2 and 3.

5.7 Qualitative Case Study

Token-level Rectification Visualization. Prompt contamination occurs when models attend to task-irrelevant background objects, biasing candidate-token aggregation and degrading retrieval. We visualize token-level attention on DomainCOCO by comparing DePro and TGR-Net heatmaps, where the rectified attention A_i reweights tokens before prompt generation.

Analysis. As shown in Fig. 9, DePro often shifts attention toward visually prominent distractors (blue boxes), allowing non-target evidence to enter prompt generation. After TGR, the rectified attention A_i becomes more target-centric and better aligned with the designated target regions (red boxes). This does not mean that TGR-Net performs explicit object detection; rather, it shows that semantic guidance can reshape candidate-token distributions before prompt generation, supporting the diagnostic view that prompt contamination is observable inside the prompt-construction pathway.

5.8 Discussion and Practical Considerations

Computational Overhead. TGR-Net adds token rectification plus optional MSF and gallery-side refinement. The student can run in

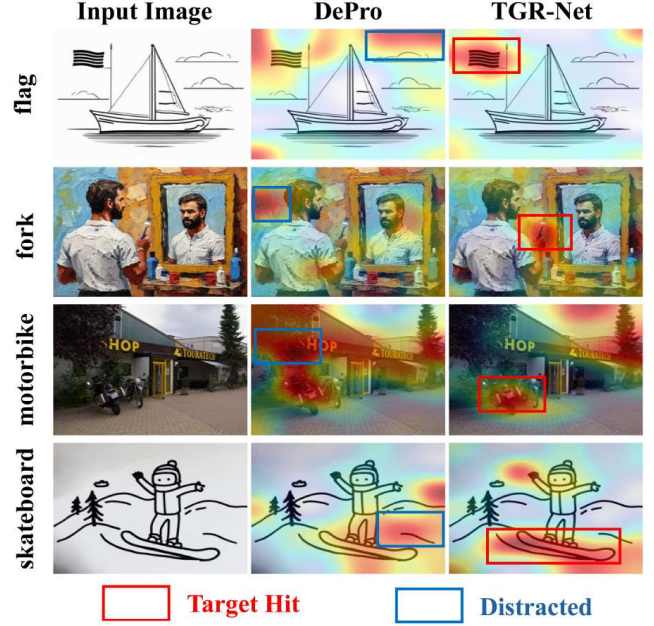


Figure 9: Visualization of token-level rectification on DomainCOCO. We show the input image (left), DePro attention (middle), and TGR-Net rectified attention (right). Red and blue boxes highlight target and distractor regions, respectively. TGR-Net shifts more attention toward the primary target and away from background clutter, helping reduce prompt contamination.

single-scale mode, and gallery refinement can be pre-computed for static galleries.

Generative Bias and Benchmark Scope. Because DomainCOCO uses SDXL-based stylization, it may inherit generator-specific artifacts. We therefore position it as a controlled diagnostic benchmark that complements naturally collected data by isolating target-aware retrieval under contextual distractors.

Relation to Detection-based Pipelines. Detect-then-encode is complementary, but introduces an external region-proposal and crop-retrieval stage. DomainCOCO instead diagnoses prompt-based UCDR directly, while TGR-Net tests candidate-token rectification without a detection-first system.

6 Conclusion

DomainCOCO exposes prompt contamination in Realistic UCDR by testing designated targets amid contextual distractors. TGR-Net provides a practical text-guided rectification baseline, with consistent gains on DomainNet and larger improvements under cluttered DomainCOCO conditions.

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GenAI Usage Disclosure

This work uses generative models as part of the research methodology: SDXL-based image generation constructs the stylized DomainCOCO domains described in Sec. 3.1. GenAI tools were also used for coding assistance and language polishing. All research ideas, experimental design, analysis, benchmark curation, and final manuscript content were determined and verified by the authors, who take full responsibility.

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